



OptiZone Startup - Semester Project

Workload Data Processing & Visualization Pipeline for ACL Injury Prevention in Women's Football

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GitHub Code Repository: github.com/Mix-ABELA/OptiZone-Data-Visualisation

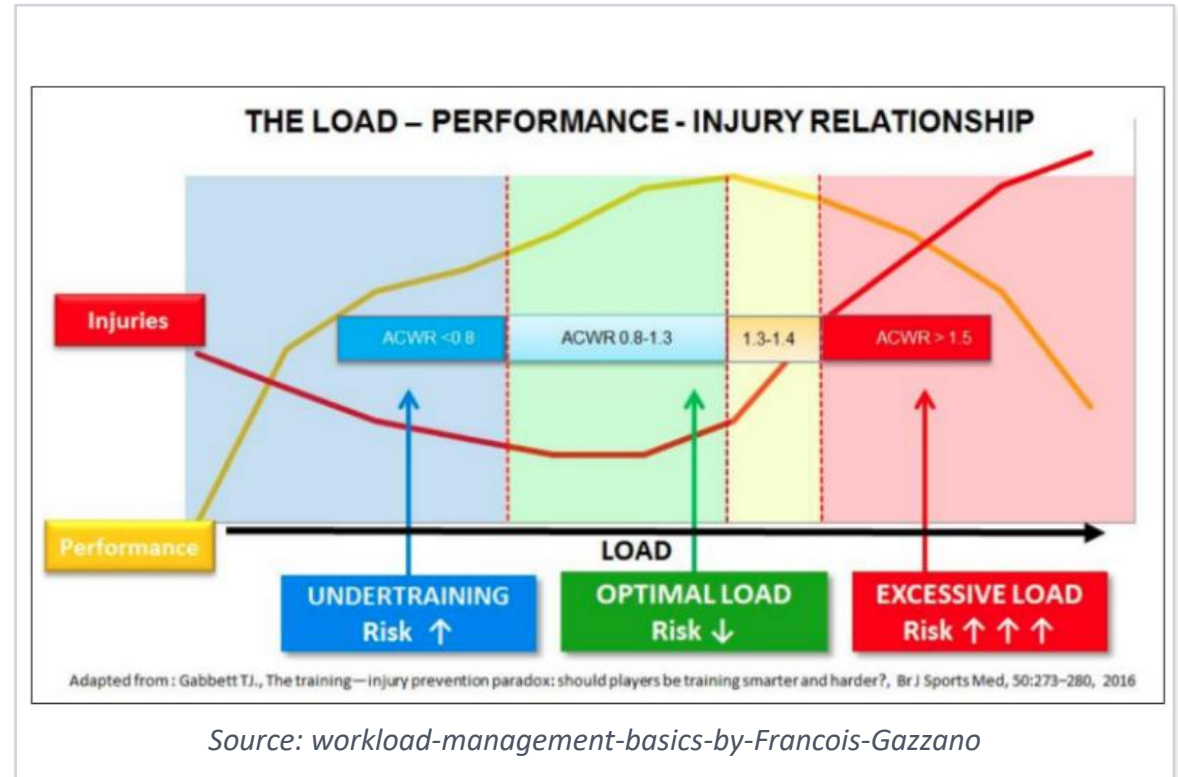


Structure of the presentation

- 1. Introduction / Context
- 2. Objectives of the project
- 3. Solution & methodology
- 4. Results (validation of achievements)
- 5. Conclusion & discussion

Why workload monitoring matters

- ACL injuries have a high burden in women's football.
- Female athletes show a higher ACL injury incidence than male athletes in similar pivoting sports.
- Monitoring external load (GPS) and internal load (Heart Rate) helps coaches manage fatigue and avoid risky load spikes.



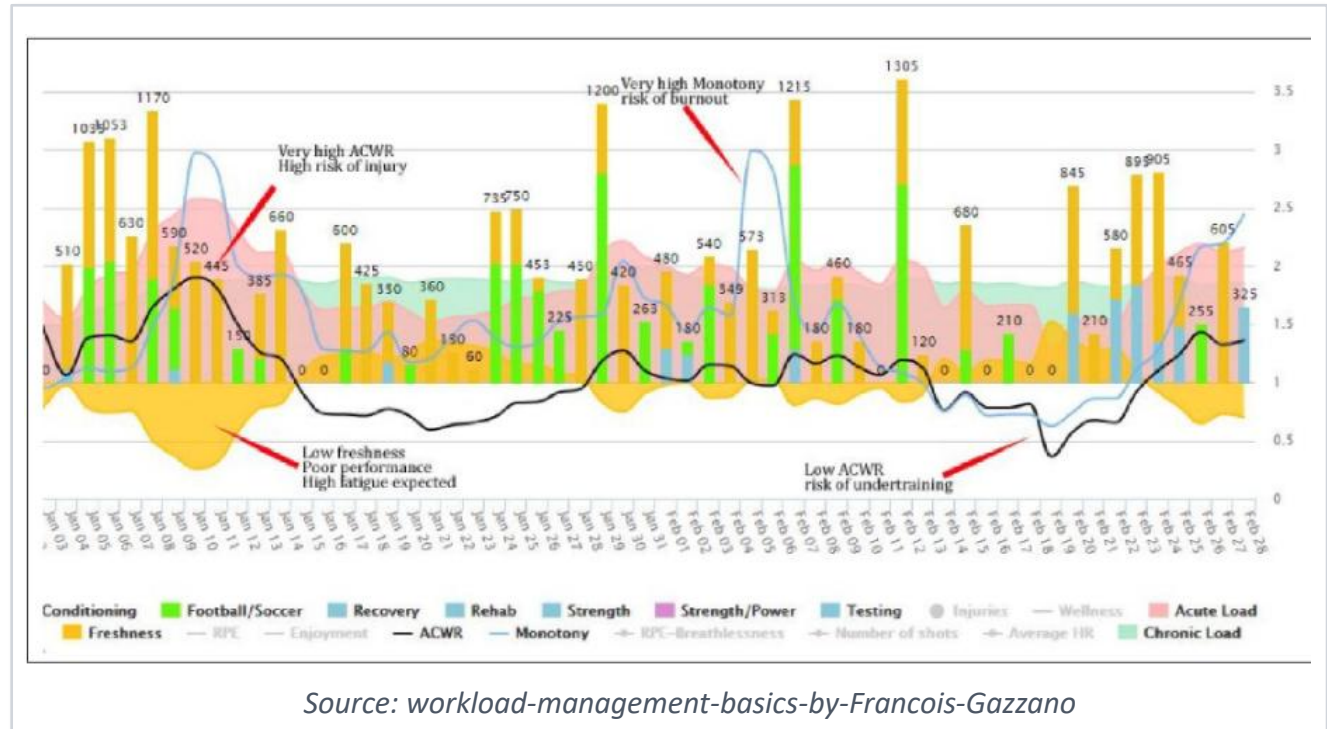
From raw data to workload decisions

Two complementary views of training/match demand:

- External load (GPS): distance, speed zones, accelerations.
- Internal load (HR): physiological response and intensity distribution.

Challenge for practitioners

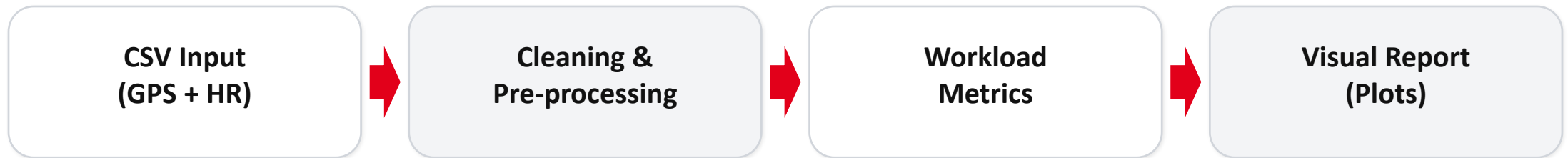
- Wearable data is noisy and device-dependent.
- Metrics must be reliable and interpretable.
- Outputs must be easy to reuse across athletes & sessions.



Project objectives

- Build an end-to-end pipeline: from raw match CSV data → cleaned signals → workload metrics → coach-oriented plots.
- Implement robust data cleaning (GPS signal quality filtering, time consistency, kinematic derivation).
- Compute key workload indicators (external + internal load) with configurable thresholds.
- Generate reusable visual reports: GPS/HR plots + pitch trajectory + heatmap.
- Design a modular, configuration-driven codebase suitable for server deployment and future scaling.

Solution overview (pipeline)



Implementation (modular handlers)

- csv_handler → load + validate + filter raw samples
- metrics_handler → compute external/internal load indicators
- plot_handler → generate GPS/HR/pitch figures for coaches
- config_handler → read config_params.ini (thresholds & toggles)

Data cleaning & processing

Goal: convert noisy wearable data into reliable kinematics (speed/acceleration) for metrics.

- CSV validation: required columns, types, and chronological ordering.
- GPS quality filters: minimum satellites + minimum signal quality (configurable).
- Kinematics: derive speed from consecutive positions; derive acceleration from speed.
- Robustness: ignore unrealistic spikes to avoid inflating acceleration-based metrics.
- Pitch mapping: convert lat/lon to a pitch reference frame using configured corner coordinates.

Workload metrics & configurability

External load (GPS)

- Total distance covered
- Instantaneous speed / pace
- Speed zones (time + %)
- Acceleration / deceleration indicators
- High-intensity actions (threshold-based)

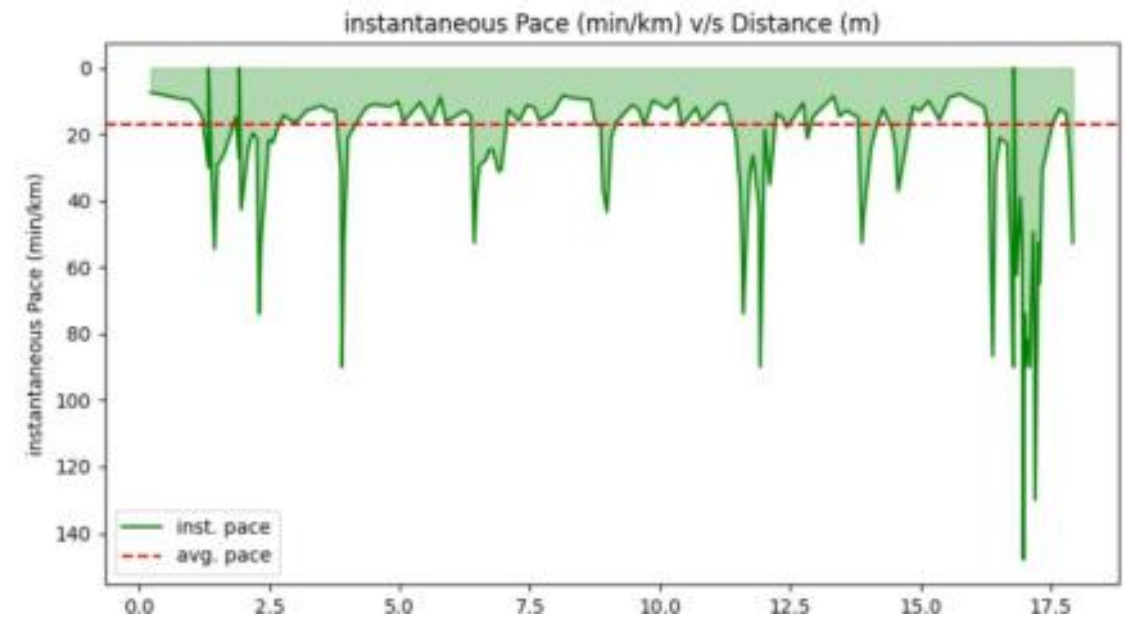
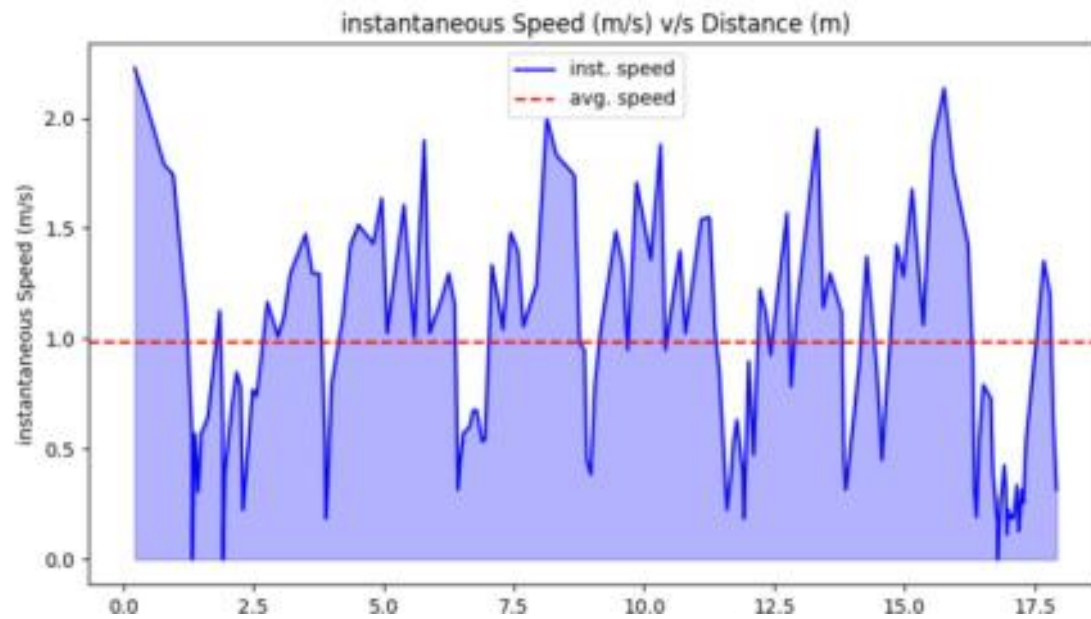
Internal load (Heart rate)

- Average & maximum HR
- HR zones distribution (%)
- Relative intensity (resting HR + age-based estimates)
- Simple efficiency indicators (optional)

Config-driven behavior (config_params.ini)

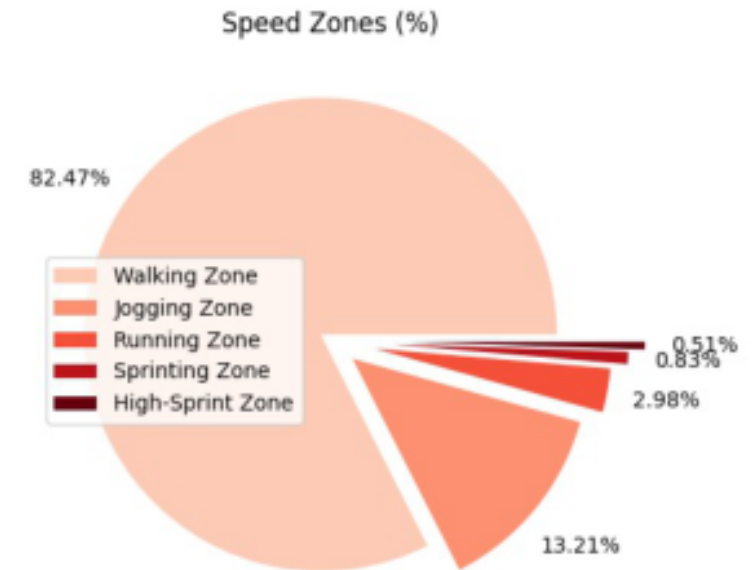
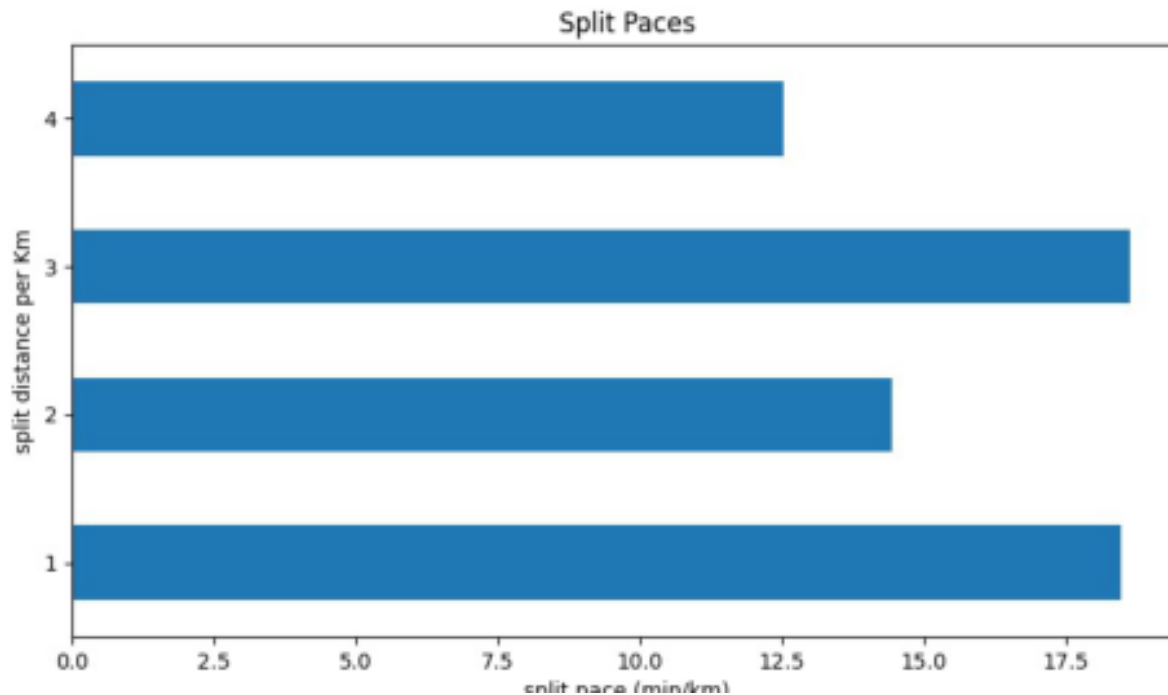
- Select dataset + enable/disable GPS/HR/pitch plots
- Set GPS quality thresholds (satellites, signal quality)
- Define speed thresholds for zones + pitch corner coordinates

Results — GPS plots (1)



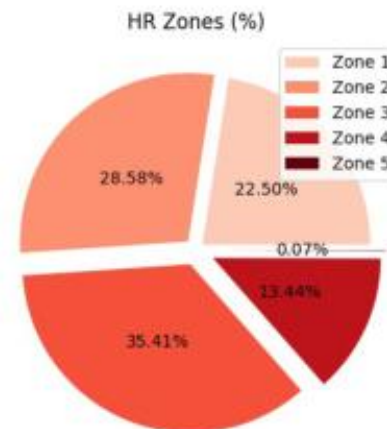
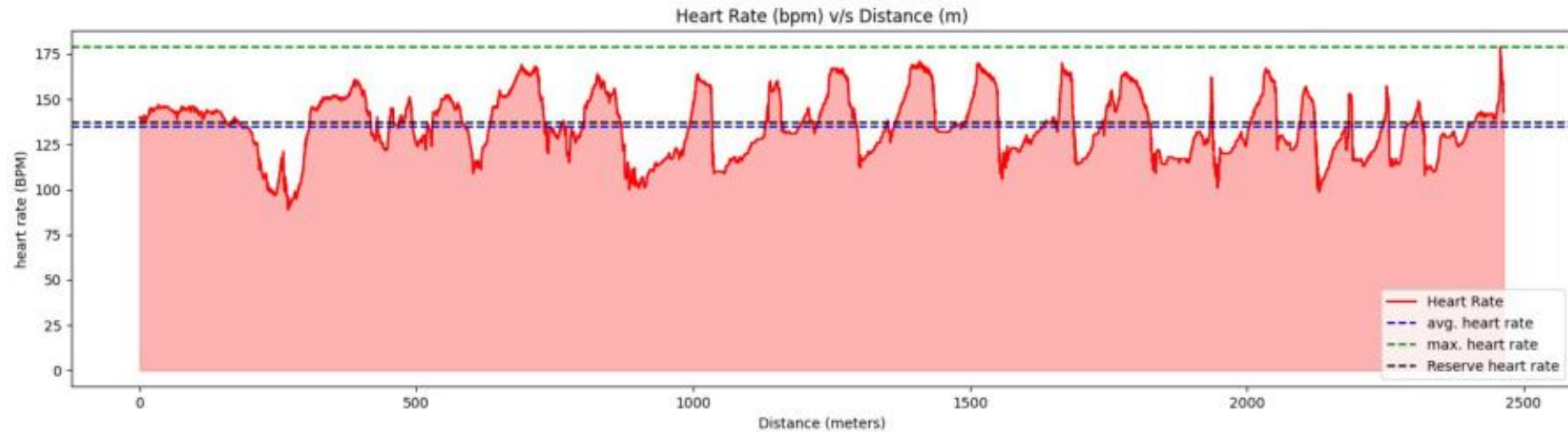
Speed & pace over distance

Results — GPS plots (2)



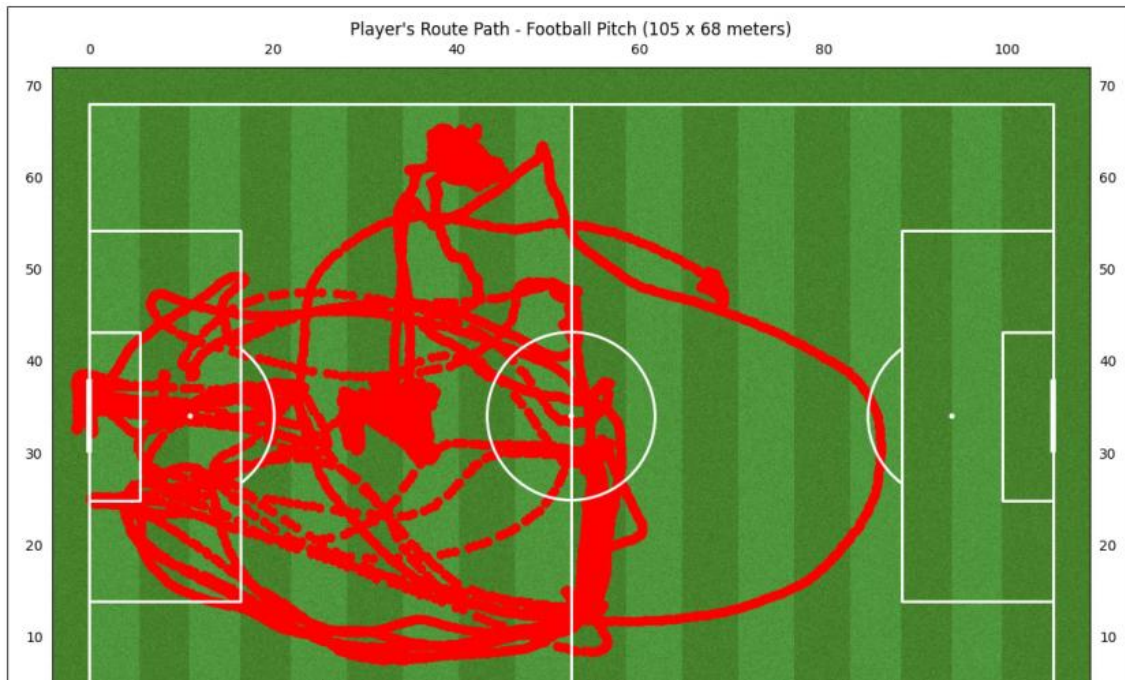
Split paces & Speed Zones distribution

Results — Heart Rate plots



Heart Rate & Intensity Distribution zones

Results — Pitch visualization plots



Player's Trajectory (where the athlete moved)



Player's Heatmap (where time was spent)

Conclusion & discussion

Validated achievements

- End-to-end pipeline from raw CSV → cleaned data → metrics → plots.
- Config-driven thresholds and toggles for reuse across athletes/sessions.
- Coach-oriented outputs: GPS/HR intensity summaries + pitch context.
- Modular codebase ready for integration into a server workflow.

Discussion

- Wearable data quality varies (signal drops, noise) → filtering is critical.
- Current validation uses one match dataset; broader evaluation requires more sessions/athletes.

Thank you for listening — Questions?

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